

## Chapter 19

# Mineral Based Industry

## Class 10 - Total Geography Morning Star

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### Choose the correct option

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#### Question 1

..... industry forms the backbone of industrial development in any country.

1. Chemical
2. Iron and steel
3. Coal
4. Electronics

#### *Answer*

Iron and steel

#### Question 2

..... is used for hardening of steel.

1. Manganese
2. Copper
3. Nickel
4. Chromium

#### *Answer*

Manganese

#### Question 3

Iron is galvanised by using chromium, ..... and tungsten.

1. copper
2. silica

3. carbon
4. nickel

**Answer**

nickel

**Question 4**

The molten iron that is obtained by the process of ore reduction is known as ..... .

1. Cast iron
2. Wrought iron
3. Pig iron
4. Steel

**Answer**

Pig iron

**Question 5**

..... is the largest steel making company in India in the Public sector.

1. Steel Production Unit of India.
2. Steel Manufacturing Unit of India.
3. Steel Company of India.
4. Steel Authority of India.

**Answer**

Steel Authority of India

**Question 6**

Which of the following are the integrated Iron and Steel plants where SAIL produces Iron?

1. Bhilai and Durgapur
2. Salem and Bhadravati
3. Visakhapatnam and Vijayanagar
4. None of the above.

**Answer**

Bhilai and Durgapur

**Question 7**

..... is the oldest steel plant in the country.

1. TISCO
2. CISCO
3. DELCO
4. TELCO

**Answer**

TISCO

**Question 8**

Bhilai Steel plant was established in collaboration with ..... in ..... .

1. Germany; West Bengal
2. erstwhile USSR; Chhattisgarh
3. France; Madhya Pradesh
4. Japan; Odisha

**Answer**

erstwhile USSR; Chhattisgarh

**Question 9**

Bhilai Steel Plant is located in Durg district of ..... .

1. West Bengal
2. Madhya Pradesh
3. Chhattisgarh
4. Odisha

**Answer**

Chhattisgarh

**Question 10**

The main source of power for the Bhilai Steel Plant comes from the power station at ..... .

1. Korba
2. Balaghat
3. Durg
4. Bokaro

**Answer**

Korba

**Question 11**

The Rourkela Steel Plant was built with cooperation from ..... in .....

1. Britain; Bhirmitrapur
2. France; Baradwar
3. Russia; Keonjhar
4. Germany; Sundargarh

**Answer**

Germany; Sundargarh

**Question 12**

Visakhapatnam is the first ..... based steel plant in India.

1. nuclear power
2. coal
3. shore
4. petroleum

**Answer**

shore

**Question 13**

Visakhapatnam steel plant obtains iron ore from ..... in Chhattisgarh.

1. Korba
2. Bailadila
3. Talcher
4. Durg

**Answer**

Bailadila

**Question 14**

..... are organic chemicals derived from petroleum products; LPG and coal.

1. Fossils fuels
2. Pharmaceuticals
3. Petrochemicals
4. Petro derrivatives

**Answer**

Petrochemicals

**Question 15**

Which of the following are mismatched?

1. Electronics: Kanpur
2. Petrochemicals: Chennai
3. Software: Bengaluru
4. Entertainment: Mumbai

**Answer**

Electronics: Kanpur

**Answer the following questions**

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**Question 1(a)**

Which is the largest mineral-based industry in India? Why is it called the backbone of all industries?

**Answer**

Iron and Steel industry is the largest mineral-based industry in India.

It is called the backbone of all industries because it provides raw material for making industrial machinery, electrical machinery, defence equipment, railway tracks, railway engines, bridges, dams, shops, automobiles, houses and a host of other industrial and consumer goods.

### Question 1(b)

What is an integrated steel plant? Give one example.

#### **Answer**

An integrated steel plant is the one in which all the processes from providing raw materials, basic fuels, water supply, etc., to the conversion of steel, rolling, etc., are all done at one place.

For example- Bhilai Iron and Steel Plant, Bhilai.

### Question 1(c)

State any three factors which affect the location of an integrated steel plant.

#### **Answer**

Three factors which affect the location of an integrated steel plant are-

1. **Availability of raw material** — Usually the integrated iron and steel plants are located near the source of raw material.
2. **Power supply** — Adequate power supply facilities are important for working with various machines.
3. **Labour force** — Availability of cheap labour force is also important.

### Question 1(d)

(i) How is it advantageous for a mini steel plant to use electric furnaces?

(ii) How is it advantageous for a mini steel plant to be located far from the source of the raw material?

#### **Answer**

(i) It is advantageous for a mini steel plant to use electric furnaces because of the following reasons-

1. Dependency on electric power does not cause pollution.
2. It conserves coking coal.
3. Less capital investment is required.

(ii) It is advantageous for a mini steel plant to be located far from the source of the raw material because of the following reasons-

1. Mini steel plants can be distributed all over the country and cater to the demands of the local market.

2. They reduce pressure on the large integrated steel plants.
3. They use scrap iron as raw material which is easily available and is comparatively cheap.

### **Question 2(a)**

Name two major steel plants in the Public Sector. Name one steel plant that was set up with Russian collaboration.

#### ***Answer***

Two major steel plants in the Public Sector are-

1. Visakhapatnam Steel Plant, Andhra Pradesh
2. Rourkela Steel Plant, Odisha

Bhilai Iron and Steel Plant was set up with Russian collaboration.

### **Question 2(b)**

What are mini steel plants?

#### ***Answer***

Mini Steel plants generally use ferrous scrap, pig iron or sponge iron as their raw material. They work through electric furnaces and have a capacity from 10,000 tonnes to about 5 lakh tonnes per year.

Their period of construction and gestation is also short. They are distributed in different parts of the country particularly in areas away from the integrated steel plants.

### **Question 2(c)**

Mention three factors which have helped Jamshedpur to develop into India's major centre of steel production.

#### ***Answer***

Three factors which have helped Jamshedpur to develop into India's major centre of steel production are-

1. **Proximity to raw material** — Jamshedpur is located close to -
  - i. the iron ore deposits in Gurumahisani mines in Mayurbhanj district, Odisha and Noamundi mines in Singhbhum district, Jharkhand
  - ii. manganese deposits from Joda in Keonjhar district
  - iii. limestone, dolomite and fire clay from Sundargarh district, Odisha

- iv. coal deposits in Jhari and Bokaro coalfields
- 2. **Cheap Labour** — Labour is employed from the densely populated valley of Ganga, mostly from the states of Bihar, West Bengal, Jharkhand, Chattisgarh and Uttar Pradesh.
- 3. **Transport Facilities** — Jamshedpur is well connected with roads and railways to the other parts of the country. Transport is facilitated by the Eastern Railways. It is also connected with the Kolkata port for exporting of finished steel.

### Question 2(d)

Name the steel plant located in Andhra Pradesh. What two advantages this steel plant have over the other steel plants in the public sector?

#### **Answer**

Visakhapatnam Steel plant is located in Andhra Pradesh.

Two advantages this steel plant has over the other steel plants in the public sector are-

1. Visakhapatnam Steel Plant is a modern integrated steel plant with the latest technology and some of the largest units.
2. Visakhapatnam's modern protected harbour and deep land locked port facilitates import of raw materials and export of finished goods.

### Question 3(a)

What are petrochemicals?

#### **Answer**

Petrochemicals are important organic chemicals derived from petroleum products, LPG and coal.

### Question 3(b)

Why are traditional raw materials being replaced by petrochemical products?

**or**

Why are petrochemical products growing in popularity?

#### **Answer**

Petrochemicals are cost effective, economically stable, cheaper as they are produced on a mass scale. Its raw material is easily available, not dependent on agricultural raw material as in the case of jute. Therefore, raw materials like wood, glass and metals are being replaced by petrochemical products.



**Question 3(c)**

- (i) Name any two products made from petrochemicals.
- (ii) Why is a petrochemical industry usually located near an oil refinery?

**Answer**

- (i) Two products made from petrochemicals are-
1. Plastic sheets
  2. Resins and adhesives for industries
- (ii) A petrochemical industry is usually located near an oil refinery which can supply its basic requirements of Naphtha or Ethylene and Benzene.

**Question 3(d)**

The electronic industry has made an impact on both media and education. Give reasons to support your answer.

**Answer**

Electronic industry, with mass scale integration process, has produced computers, servers, displays, televisions and cameras, telephones etc. These devices help to capture and broadcast news, advertisements, cinema, educational programs etc., to a large section of the population over the country and overseas, thus changing the life style of the Indian masses.

Various new technologies like smart boards, projectors, video clips, pictures etc., are used to make learning easy and fun for the students. Further, in this Corona crisis, online classes provided to the students by the use of various online applications have helped the students to continue their education from home.

Thus, the electronic industry has made an impact on both media and education.

**Question 4(a)**

Name two industrial towns which are connected with the production of iron and steel.

**Answer**

Two industrial towns which are connected with the production of iron and steel are-

1. Visakhapatnam
2. Jamshedpur

**Question 4(b)**

Mini steel plants cause less pollution than integrated steel plants. Give reasons.

**Answer**

Mini steel plants cause less pollution than integrated steel plants because mini steel plants use electric furnaces while the integrated steel plants use coal to melt iron ore.

**Question 4(c)**

State the difficulties experienced by the iron and steel industry in India.

**Answer**

The difficulties experienced by the iron and steel industry in India are-

1. **Capital intensive** — Huge amount of capital is required by the iron and steel industry and the plants have a long gestation period.
2. **Obsolete technology** — The machinery being used in the iron and steel plants is outdated.
3. **Limited availability of Coking coal** — The coal reserves of India are limited. Many steel plants import coking coal.
4. **Sick industrial units** — Due to inadequate supply of power and sharp increase in the cost of raw material, many steel plants are experiencing sickness or have been closed down.
5. **Control of Prices** — The government has fixed the price for iron and steel which leaves little margin of profit for the manufacturers.
6. **Inefficiently managed public sector** — Many public sector plants are facing strikes, lockouts, energy crisis caused by heavy investment on social overheads, poor labour relations, inefficient management and underutilisation of capacity.
7. **Heavy demand** — To meet the increasing demand, large quantity of iron and steel has to be imported.

**Question 4(d)**

Give three reasons why the iron and steel industry is concentrated largely in the Chota Nagpur belt.

**Answer**

The iron and steel industry is concentrated largely in the Chota Nagpur belt because of the following reasons-

1. Iron and steel industry requires iron ore, coal, limestone and manganese which are available in close proximity.

2. Cheap labour is available to the industries from the states of West Bengal, Bihar, Jharkhand, Odisha and Chhattisgarh.
3. Good network of roads and railway facilitates the movement of raw materials to iron and steel plants.

## Thinking Skills

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### Question 1

Trace the historical events that led to the establishment of TISCO. List the factors that facilitated Sir Jamshedji Tata to set up the Steel Plant. Enumerate the benefits that India had due to this steel plant.

### Answer

Sir Jamshedji Tata recognized the importance of steel in India's development and desired to establish a steel plant in the country. He acquired mining rights for rich iron ore deposits in the Chota Nagpur region and acquired land in Jamshedpur in Jharkhand, where the steel plant was established.

The factors that facilitated Sir Jamshedji Tata to set up the steel plant include the following:

1. Location — Jamshedpur is most ideally located with respect to iron ore, fuel and flux supplies.
2. Availability of Raw Materials — The plant obtains its requirements of iron ore from Gurumahisani mines, Odisha and from the Noamundi mines, Jharkhand. Manganese from Joda in Keonjhar district and limestone, dolomite and fire-clay from Sundargarh district, Odisha. Coal is obtained from the Jharia and Bokaro coalfields.
3. Power Supply — Coal is the main source of power for this plant.
4. Water Supply — Rivers Kharkai and Subarnarekha supply a continuous stream of water for cooling purposes.
5. Labour Force — The labour force is recruited from the densely populated valley of Ganga.
6. Market — The plant is located at a distance of 240 km from Kolkata.
7. Transport Facilities — Jamshedpur is well connected with roads and railways to the other parts of the country. It is also connected with the Kolkata port for exporting of finished steel.

The establishment of TISCO brought several benefits to India:

1. Industrialisation and Economic Growth
2. Infrastructure Development
3. Technological Advancement
4. Self-Reliance in Steel Production

5. Diversification of Industries
6. Social Development for the local community

## **Question 2**

If you are given a choice to become an Electronics Engineer or a Software Engineer, which one you would choose and why?

### ***Answer***

I would prefer to become a Software engineer due to the following reasons:

1. High Demand — The field of software engineering offers a wide range of job opportunities.
2. Lucrative Salaries — Software engineers often enjoy competitive salaries and attractive benefits.
3. Flexibility and Work-Life Balance — Many software engineers have the option to work remotely and enjoy flexible work hours, providing a better work-life balance.
4. Creativity and Problem-Solving — Software engineering involves creative thinking and problem-solving, allowing individuals to develop innovative solutions and contribute to technological advancements.
5. Continuous Learning — The dynamic nature of technology requires software engineers to continuously learn and adapt to new programming languages, frameworks and tools.
6. Entrepreneurial Opportunities — Software engineering provides a foundation for entrepreneurial ventures.
7. Collaboration and Teamwork — Software engineers often work in collaborative teams, fostering a sense of camaraderie and the opportunity to work with diverse professionals.
8. Global Job Market — Software engineering skills are in demand globally.